FPPF TORSORS UNDER A SMOOTH COMMUTATIVE GROUP ALGEBRAIC SPACE OVER A FIELD ARE TORSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The author leaves one verification to the reader ('the reader can verify that the scheme on the left has a K-rational point'); recorded as author's own style, content verbatim, no rewrite.

Editorial note. The label lemma-torsors-over-field-torsion names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- fppf cohomology H^1 and G-torsors over Spec(K) for G a smooth commutative group algebraic space over a field K
- the torsor [n](P) whose class is n times the class of P in H^1_fppf(Spec(K), G)
- smooth schemes/spaces have points over finite separable extensions; quotients by finite constant groups

Lemma 0.1. *Let G be a smooth commutative group algebraic space over a field K. Then $H^1_{fppf}(\mathrm{Spec}(K), G)$ is torsion.*

Proof. Every G-torsor P over $\mathrm{Spec}(K)$ is smooth over K as a form of G. Hence P has a point over a finite separable extension L/K. Say $[L : K] = n$. Let $[n](P)$ denote the G-torsor whose class is n times the class of P in $H^1_{fppf}(\mathrm{Spec}(K), G)$. There is a canonical morphism

$$P \times_{\mathrm{Spec}(K)} \ldots \times_{\mathrm{Spec}(K)} P \to [n](P)$$

of algebraic spaces over K. This morphism is symmetric as G is abelian. Hence it factors through the quotient

$$(P \times_{\mathrm{Spec}(K)} \ldots \times_{\mathrm{Spec}(K)} P)/S_n$$

On the other hand, the morphism $\mathrm{Spec}(L) \to P$ defines a morphism

$$(\mathrm{Spec}(L) \times_{\mathrm{Spec}(K)} \ldots \times_{\mathrm{Spec}(K)} \mathrm{Spec}(L))/S_n \longrightarrow (P \times_{\mathrm{Spec}(K)} \ldots \times_{\mathrm{Spec}(K)} P)/S_n$$

and the reader can verify that the scheme on the left has a K-rational point. Thus we see that $[n](P)$ is the trivial torsor. □

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